SRI LANKA INSTITUTE OF INFORMATION - TECHNOLOGY

FACULTY OF COMPUTING

IE2022 - Introduction to Cyber Security

Year 2 Semester 1

IT22121592

HERATH H.M.C.H.K

Zero-Trust Architecture.

- Abstract:

As cyber threats become more advanced and frequent the traditional security methods relying on perimeter defenses are proving to be insufficient. To address this companies are embracing Zero Trust Architecture known as ZTA as a security strategy. This document explores the basics of ZTA. How it is integrated into infrastructures. It discusses the principles of ZTA highlighting the significance of verification and strict access controls. Additionally, it examines elements of ZTA such, as micro-segmentation least privilege access, and thorough identity management. Through case studies and real-world scenarios, successful ZTA implementations in industries are showcased to demonstrate the effectiveness of this approach in managing cyber risks. Furthermore, common challenges and considerations in adopting ZTA are addressed, including shifts in culture and complexities in integration. In summary, this document offers insights, for organizations aiming to strengthen their cybersecurity stance by embracing Zero Trust Architecture.

- Introduction to the topic:

Zero trust is a cyber security model that assumes that no one/ no connection can be trusted, it is centered around the idea that an organization should not automatically trust anyone or anything that is inside or outside automatically, without verifying.

In today's world where everything is connected the fashioned security methods that rely on boundaries are no longer enough to protect against increasingly complex cyber threats. With businesses depending more on cloud services, mobile devices, and remote working setups there is a growing need, for a more flexible security framework. This is where Zero Trust Architecture (ZTA) comes into play. It's a way of thinking about cybersecurity that challenges the idea of trusting entities just based on where they are in the network.

Zero Trust Architecture (ZTA) is about "Never trust, Always verifies." Unlike security strategies that trust users based on their network location, ZTA focuses on verifying trust regardless of where users are or how they connect to resources. This method fits well with today's threat

landscape where attackers often hide within trusted networks or use stolen credentials to access information.

Report Goals;

This report aims to explain Zero Trust Architecture (ZTA) and its importance, in cybersecurity practices. The key objectives of this report include;

- Clarifying the principles of Zero Trust Architecture (ZTA)
- Exploring the elements and structure of Zero Trust Architecture (ZTA, including micro-segmentation least privilege access, and identity-based controls.
- Analyzing real-life examples and case studies showcasing implementations across various industries.
- Examining the obstacles and factors involved in adopting Zero Trust Architecture, such as cultural changes and integration complexities.
- Offering insights into the research and advancements in Zero Trust Architecture encompassing emerging technologies and recommended practices.

Delving into the Topic;

Zero Trust Architecture (ZTA) signifies a shift in cybersecurity strategy by prioritizing the security of assets and transactions over-reliance on network boundaries. At its essence, ZTA operates on the principle of "never trust, always verify " necessitating continual authentication and authorization for users and devices seeking resource access.

A crucial component of ZTA is micro-segmentation which entails partitioning the network into isolated segments to restrict attackers' lateral movement. This meticulous approach to network segmentation aids, in containing breaches and reducing security incident impacts.

Another vital element of ZTA is the least privilege access principle, which limits user access rights to resources needed for their job tasks. By using privileged access controls companies can lower the chances of insider threats. Minimize the potential harm caused by compromised login details.

Identity-based measures are crucial, in Zero Trust Architecture (ZTA) as they allow organizations to confirm the identity and reliability of users and devices trying to access their network resources. This might entail using methods like factor authentication (MFA) biometric checks or other ways to verify identities to ensure that only authorized individuals can access sensitive information.

Real-life examples and case studies demonstrate how Zero Trust Architecture effectively reduces cyber risks and improves security stance. Companies in sectors such as finance, healthcare, and government have implemented ZTA principles successfully to safeguard their assets and confidential data from unauthorized access or breaches.

Although there are advantages to embracing Zero Trust Architecture organizations face hurdles. Resistance to change within company culture limitations posed by existing infrastructure and difficulties in integrating security technologies are challenges that need addressing. Nonetheless, through planning cooperation among teams and investment in cutting-edge cybersecurity solutions companies can enjoy the benefits of a security approach based on Zero Trust principles.

In summary, Zero Trust Architecture (ZTA) signifies a change in cybersecurity practices by providing a proactive and adaptable method, for guarding against emerging threats.

By adopting verification principles implementing micro-segmentation and least privilege access well and using identity-based controls companies can enhance their security stance and safeguard their important assets in the ever-changing landscape of threats today.

- Evolution of the topic:

> *" Zero trust (ZT) provides a collection of concepts and ideas designed to minimize uncertainty in enforcing accurate, least privilege per-request access decisions in information systems and services in the face of a network viewed as compromised. Zero trust architecture (ZTA) is an enterprise's cybersecurity plan that utilizes zero trust concepts and encompasses component relationships, workflow planning, and access policies. Therefore, a zero trust enterprise is the network infrastructure (physical and virtual) and operational policies that are in place for an enterprise as a product of a zero trust architecture plan "*

History of zero trust architecture

The concept of Zero Trust Architecture (ZTA) has its origins, in the 2000s emerging as a response to the changing landscape of cybersecurity and the limitations of perimeter security models. Although the term "Zero Trust" was coined later its foundational principles draw from cybersecurity ideologies and concepts.

1. Early Influences (the 2000s);

Jericho Forum- During the 2004s the Jericho Forum, a community of IT security experts started advocating for a new cybersecurity approach centered around de-parameterization. They argued that traditional network boundaries were becoming obsolete due to advancements like cloud computing, mobile technology, and remote work.

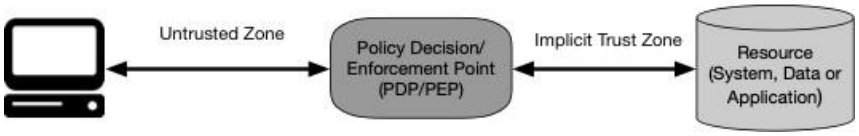
BeyondCorp -The Google BeyondCorp initiative, launched around 2011 marked a shift away from VPN-centric access to internal systems. Instead, it promoted a model where access privileges were determined by user identity and device security status, rather than network location.

2. The Emergence of Zero Trust (2010s)

Forrester Research released a report, in 2010 called "No Chewy Centers; Introducing the Zero Trust Model of Information Security " which helped popularize the Zero Trust concept. The report suggested that organizations should not automatically trust anything inside or outside their boundaries but instead should verify any connection attempts before granting access.

The Zero Trust model gained momentum in the cybersecurity community with influential figures like John Kindervag from Forrester Research advocating for its adoption. Kindervag highlighted principles of Zero Trust emphasizing authentication least privilege access and strict access controls.

In 2020 the National Institute of Standards and Technology (NIST) issued Special Publication 800- 207 "Zero Trust Architecture " offering guidance on implementing Zero Trust principles in agency networks. This publication from NIST reinforced the credibility and significance of Zero Trust, in cybersecurity practices.



Industry adoption Standardization (2010s Present)- Major corporations, in industries have started adopting Zero Trust principles to enhance their cybersecurity measures. They realized the shortcomings of perimeter-based security. Sought more effective ways to defend against advanced cyber threats.

Leading industry groups and standards bodies like the Cloud Security Alliance (CSA) and the International Organization for Standardization (ISO) have begun integrating Zero Trust concepts into their frameworks and guidelines. This move aims to establish terminology and best practices for implementing zero-trust security protocols.

In terms of progress from the 2010s -

Organizations have leveraged advancements in network segmentation technologies to implement zero-trust strategies by dividing networks into smaller isolated segments. This strategy helps prevent movement by cyber attackers and minimizes the impact of security breaches.

Identity and Access Management (IAM) solutions have evolved to support Zero Trust by offering authentication and authorization mechanisms based on user identity and context. Features like factor authentication (MFA) biometric authentication and continuous monitoring have become essential components of Zero Trust architectures.

The emergence of Zero Trust Network Access (ZTNA) has also played a role, in enhancing organizations' cybersecurity postures.

ZTNA solutions have been developed to meet the needs of mobile workers providing access, to applications and resources without relying on traditional VPNs. These solutions prioritize user authentication and access controls at the application level.

Recent advancements and prospects;

Incorporating AI and Automation; The integration of intelligence (AI) and automation technologies is projected to improve Zero Trust architectures by enabling real-time threat detection adaptive access controls and automated response mechanisms.

Ongoing Progression; Zero Trust Architecture is continuously evolving in response to emerging threats, progress, and evolving business demands. Organizations are anticipated to enhance their Zero Trust strategies and implement solutions for evolving cybersecurity challenges.

In essence, the evolution of Zero Trust Architecture signifies a shift in cybersecurity mindset moving away from the concept of perimeter-based security, towards a dynamic approach centered on continual verification and stringent access controls. As organizations grapple with cyber threats Zero Trust Architecture is expected to remain a crucial element of contemporary cybersecurity strategies.

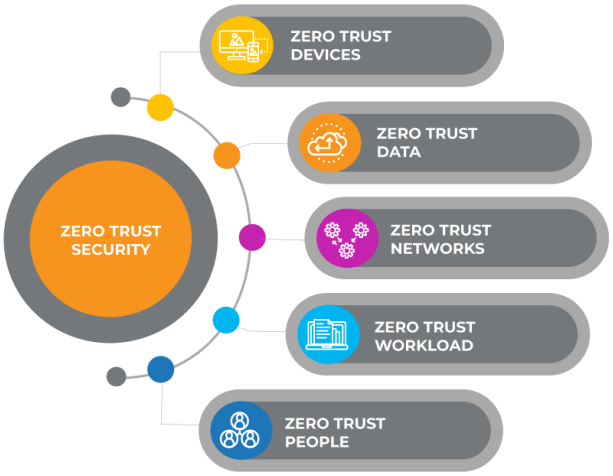
The Principles of Zero Trust Architecture

Zero Trust Architecture (ZTA) is built upon several core principles that fundamentally challenge traditional security models based on perimeter defenses and implicit trust. These principles collectively form the foundation of a proactive and adaptive approach to cybersecurity, emphasizing continuous verification, least privilege access, and strict access controls.

1. Never Trust, Always Verify –

   This principle is the cornerstone of Zero Trust Architecture, the bedrock of the ZTA. Unlike security approaches that rely on perimeter defenses and trust based on network location Zero Trust advocates for verifying trust for every access request regardless of the user's location or network origin.

   Simply, Just because someone on your corporate networks says they are who they are or carrying an employee batch we can not trust them nor doesn't mean they are who they say they are. So this principle always verifies who they are.



2. Least privilege access –

   The principle of least privilege access ensures that users and devices are only given access, to the resources they need to carry out their tasks – nothing more. By limiting access rights to the minimum required organizations can reduce their vulnerability to security breaches. Minimize unauthorized access risks.

3. Micro-segmentation –

   Micro-segmentation entails dividing the network into sections or areas each, with its own specific access controls and security regulations. This detailed approach to network division enables organizations to establish security boundaries between network components thereby confining the spread of threats and restricting lateral movement by potential intruders. Micro-segmentation serves as a deterrent against intruders freely navigating the network and reaching assets once they breach it.

4. Continuous monitoring and Analysis –
   Zero Trust Architecture underscores the significance of surveillance and analysis to identify behavior and potential security risks in real-time. By gathering and scrutinizing data from origins, such as network flow, user actions, and endpoint operations organizations can promptly detect activities, unauthorized login attempts, and other security breaches. Continuous surveillance empowers organizations to promptly address emerging threats. Proactively implement measures to reduce risks.

5. Encryption everywhere –
   The concept of encryption ubiquity promotes the deployment of encryption for safeguarding data during transmission as well as when at rest. This involves encrypting communication pathways, like network flow and email transfers alongside encrypting data stored on servers, databases, and other storage mediums.

   When organizations encrypt data from the end, to end they can protect information from being intercepted listened in on, or accessed without permission even if the outer defenses are breached.

6. Secure Access controls –

The Zero Trust Architecture focuses on setting up access controls based on user identity, device status, and contextual elements. This might include using factor authentication (MFA) biometric verification checking device health and analyzing user behavior to confirm the legitimacy of access requests and ensure that only authorized users and devices can access resources. By implementing access controls unauthorized entry is. Safeguards are put in place against credential theft and misuse.

In essence, these principles collectively outline the beliefs of Zero Trust Architecture. Help organizations develop strong and adaptable cybersecurity defenses. By adopting these principles organizations can establish a security stance that emphasizes verification processes to reduce vulnerabilities to attacks and boost protection, against various cyber threats.

The element and structure of Zero Trust Architecture.

Zero Trust Architecture (ZTA) is a cybersecurity framework that challenges the approach of relying on a perimeter. It acknowledges the possibility of threats, from both inside and outside the network. In distinguishing between trusted less trusted external networks ZTA treats all network traffic as potentially harmful regardless of its origin. This strategy aims to bolster security through access controls, continuous monitoring, and robust authentication measures. The key components and framework of Zero Trust Architecture can be summarized as follows,

1. Core Principles of Zero Trust –

As I said earlier, ZTA is built on principles that shape its implementation like,

- Verify explicitly - Every user and device seeking access to resources must undergo authentication and authorization based on predefined policies.
- Least Privilege Access- Users and devices should only have access, to the level required for their tasks.
- Assume breach - Rather than trusting entities based on their network location ZTA operates under the assumption that the network is already compromised or could be compromised at any point.
- Micro-segmentation - Network segmentation occurs at a detailed level to isolate and safeguard critical assets and data.

- Continous monitoring - ZTA consistently keeps an eye on network traffic, user actions, and devices to detect any behavior or policy breaches.

2. Identify and Access management -

In ZTA IAM plays a role by offering ways to verify the identity and grant access to users and devices.

The process of verifying identity usually involves factors like passwords, biometrics, and security tokens.

Access control policies are implemented based on user roles, attributes, and contextual details to ensure that authorized individuals can access specific resources.

3. Network segmentation-

ZTA suggests dividing the network into segments to contain breaches and prevent attackers from moving laterally.

This division can be achieved through technologies like LANs (VLANs) firewalls and softwaredefined networking (SDN).

Micro-segmentation is an aspect of ZTA where the network is divided into segments with specific access controls for each segment.

4. Encryption of Data-

Encryption is highly emphasized in ZTA to safeguard data whether it is in transit or, at rest. To keep data secure while it moves across networks many organizations use Transport Layer Security (TLS) and Secure Sockets Layer (SSL) protocols, for encryption.

It's important to encrypt data stored on servers, databases, and other storage devices using methods to prevent unauthorized access in case of a security breach.

5. Continuous monitoring and analytics -

Zero Trust Architecture (ZTA) relies on monitoring and analysis to identify activities, spot potential threats, and respond quickly to security issues.

Systems like Security Information and Event Management (SIEM) intrusion

detection/prevention systems (IDS/IPS) and user behavior analytics (UBA) are vital parts of ZTAs monitoring setup.

By analyzing network traffic, logs, and user behaviors in time security teams can swiftly. Address security risks.

6. Policy Enforcement points (PEPs) -

Policy Enforcement Points (PEPs) are enforcement tools scattered across the network to uphold access control rules and check traffic compliance.

These enforcement points can include firewalls, gateways, routers, switches, and endpoint security solutions.

PEPs play a pivotal role in applying Zero Trust principles by ensuring that only authorized traffic flows, between different network segments and resources.

7. Continuous Authentication-

Continuous authentication focuses on verifying users throughout their sessions than, during login. This approach involves using methods like session tokens, biometric authentication, and behavioral analytics to monitor user activity and adjust access privileges based on potential risks.

8. API security -

API security is crucial in the Zero Trust Architecture (ZTA) framework as organizations rely more on APIs to connect systems. ZTA recommends implementing authentication and authorization measures for APIs along with encrypting data exchanged through them. Additionally deploying API gateways and security solutions helps enforce policies and safeguard against API-related security risks.

9-Secure remote access -

Ensuring access is vital in today's landscape of remote work and cloud-based services to allow users to securely access corporate resources from anywhere. ZTA utilizes technologies such, as networks (VPNs) SSL VPNs, and SASE solutions to provide secure remote access while adhering to Zero Trust principles.

10- Cloud security -

While cloud computing poses security challenges the principles of Zero Trust Architecture (ZTA) remain relevant, in cloud settings.

Companies that embrace ZTA in the cloud extend their security measures to cover cloud-based assets. They deploy identity-based access controls, encryption protocols, and ongoing monitoring to safeguard data and workloads stored in the cloud.

To sum up, Zero Trust Architecture serves as a cybersecurity framework that aims to bolster security by assuming the presence of threats both within and outside the network perimeter. Through the enforcement of access controls, continuous monitoring practices and robust authentication mechanisms organizations can reduce security vulnerabilities. Enhance protection for their critical assets and data, against cyber threats.

Analyzing real-life examples and case studies showcasing implementations across various industries.

When we examine real-world instances and case studies of Zero Trust Architecture (ZTA) implementations, in sectors we gain insights into how this cybersecurity framework is applied and the advantages it offers. Let's delve into a few examples;

1**. Googles BeyondCorp**-

- Industry - Technology
- Overview- BeyondCorp represents Google's interpretation of Zero Trust Architecture aimed at ensuring access to applications and resources for employees working remotely eliminating the need for a conventional VPN.
- Key Features- BeyondCorp shifts away from the perimeter-focused security model to a user strategy, where access decisions consider identity, device security status, and contextual elements. Employees must. Authorize themselves before accessing resources irrespective of their location.
- Impact - At Google, BeyondCorp has bolstered security levels by decreasing reliance on VPNs and perimeter defenses. Additionally, it has enhanced user experience by enabling secure resource access, from any location, device, or network.

2. **U.S. Department of Defense (DoD) zero trust architecture -**

- Industry -Government and Military
- Overview - The U.S. Department of Defense (DoD) is currently rolling out Zero Trust Architecture across its networks and systems to boost cybersecurity and safeguard data from cyber threats.
- Key Features -The DoD approach, to Zero Trust Architecture, focuses on verification of trust granting access privileges and implementing micro-segmentation to enforce strict access controls and restrict lateral movement by potential attackers. It also incorporates authentication methods and encryption technologies to secure data both in transit and, at rest.
- Impact- Through the adoption of Zero Trust Architecture the DoD aims to fortify its defenses against cyber threats lower the chances of data breaches and unauthorized access and ensure the confidentiality, integrity, and availability of information.

3. **Gartners's Zero Trust Network Access (ZTNA) Magic Quadrant**-

- Industry- Cybersecurity/Consulting
- Overview - Gartner's assessment of the Zero Trust Network Access (ZTNA) Magic Quadrant identifies vendors in the ZTNA market showcasing their capabilities in delivering remote access solutions rooted in Zero Trust principles.

- Key features - Zero Trust Network Access (ZTNA) solutions allow companies to provide access, to applications and resources based on user identity, device security status, and contextual factors eliminating the need for a VPN. These solutions typically include access controls, multi-factor authentication, and encryption to safeguard data and thwart entry.

- Impact - Businesses across sectors depend on Gartners ZTNA Magic Quadrant to assess and choose ZTNA vendors that align best with their security needs and operational requirements. Through the deployment of ZTNA solutions, organizations can boost the productivity of workers enhance their security stance, and reduce exposure to cyber risks.

4. **Adoption of Zero Trust in Healthcare** -

- Industry -Healthcare

- Brief Overview- With the growing digitalization of records and healthcare infrastructure the healthcare sector is embracing Zero Trust Architecture as a means to safeguard patient information and adhere to regulatory mandates like HIPAA.
- Key features- Healthcare entities are integrating Zero Trust principles such as trust verification least privilege access control and encryption to protect data against unauthorized access. They are also incorporating identity and access management (IAM) tools alongside analytics, for security measures and early detection of suspicious activities.

- Impact - Zero Trust Architecture has proven to be beneficial, for healthcare institutions by bolstering their cybersecurity measures ensuring the safety of data from breaches, and upholding compliance with regulations. Moreover, it has boosted confidence in the security and confidentiality of their health records.

These real-world instances and success stories showcase the ranging advantages of Zero Trust Architecture in sectors illustrating its effectiveness, in improving cybersecurity safeguarding sensitive information, and reducing the risk of cyber attacks. As businesses increasingly embrace principles and technologies they can effectively navigate evolving security threats and maintain the resilience and protection of their digital assets.

Offering insights into the research and advancements in Zero Trust Architecture encompasses emerging technologies and recommended practices.

Zero Trust Architecture (ZTA) research and progress are pivotal, in driving innovation enhancing security measures, and tackling evolving cyber threats. As businesses embrace principles experts are delving into technologies and best practices to boost the efficiency and scalability of ZTA implementations. Here's a glimpse into the developments in Zero Trust Architecture;

1. Emerging Technologies-

a. Zero Trust Network Access (ZTNA) - ZTNA solutions are gaining prominence as a technology for deploying Zero Trust Architecture in remote and hybrid work settings.

These solutions offer secure user-focused access to applications and resources of the user's location or network connection. Researchers are exploring enhancements in ZTNA technologies like software-defined perimeter (SDP) architectures and cloud-based access brokers to elevate scalability, performance, and user experience.

b. Artificial Intelligence (AI) and Machine Learning (ML) - AI and ML tools are increasingly being integrated into Zero Trust Architecture to improve threat detection, user authentication, and access control mechanisms. Researchers are working on security analytics platforms driven by AI to analyze volumes of data, in real-time for detecting unusual behavior and potential security risks. By using AI and ML algorithms businesses can enhance the precision and efficiency of their security processes while lessening the workload on analysts.

c. Blockchain technology - Blockchain technology is being investigated as a facilitator of Zero Trust Architecture for IAM applications. Identity solutions based on offer decentralized, tamper identity verification removing the necessity for centralized authorities and reducing the chances of identity theft and fraud. Researchers are exploring blockchain applications for user authentication, credential management, and identity federation within zero-trust environments.

d. Software-defined networking -SDN technologies are utilized to implement micro-segmentation and network isolation within Zero Trust Architecture. SDN enables organizations to dynamically create and manage network segments based on user identity, device posture, and application needs. Researchers are looking into SDN-based methods to automate policy enforcement streamline network traffic and improve visibility and control, in Zero Trust environments.

e. By exchanging information, on threats with trusted allies and colleagues in the industry companies can improve their ability to recognize and address security risks boost their understanding of current situations, and fortify their joint defense against cyber assaults.

In essence, continuous exploration and progress, in Zero Trust Architecture are propelling innovation. Shaping the future of cybersecurity. By utilizing technologies and recommended strategies businesses can enhance the efficiency, scalability, and robustness of their Zero Trust

setups. This empowers them to better respond to changing threats while safeguarding their assets and resources.

- Future developments in the area

Zero Trust Coordination and Automation-

Background- Studies show a rising interest, in using coordination and automation tools to streamline the implementation of Zero Trust and enhance efficiency.

Future development - Automated coordination platforms will be developed to simplify the setup, configuration, and handling of Zero Trust measures and regulations across environments. These platforms will utilize machine learning techniques and AI-based analysis to adjust security policies dynamically based on threat intelligence and user actions.

Zero Trust for Internet of Things (IoT) and Operational Technology (OT)-

Background -With the increase in devices and the merging of IT and OT networks there is a crucial need to expand Zero Trust principles to safeguard vital infrastructure and industrial systems.

Future Development - Future investigations will concentrate on crafting Zero Trust frameworks and structures tailored specifically for the demands of OT settings. This involves implementing micro-segmentation access controls based on identity and encryption to protect devices, sensors, well as industrial control systems from cyber risks.

Zero Trust in Cloud Native Environments-

Background - With organizations transitioning to native designs while embracing containerized applications there is a demand, for Zero Trust solutions that can deliver consistent security across various cloud environments including multi-cloud setups.

Future development - In the coming years, researchers will delve into strategies, for incorporating Zero Trust principles into native platforms like Kubernetes and serverless computing frameworks. This involves creating security controls centered around containers, measures during runtime, and access policies based on APIs to safeguard cloud workloads and microservices.

Zero trust for DevSecOps and Continuous integration-

Background -When it comes to DevSecOps and Continuous Integration/Continuous Deployment (CI/CD) the background lies in the emphasis on integrating security throughout software development lifecycles. However traditional security models often struggle to match the speed and agility of CI/CD pipelines.

Future Development - Looking ahead future research will concentrate on merging Zero Trust principles with DevSecOps workflows to facilitate automated security testing, vulnerability scanning, and policy enforcement across the CI/CD pipeline. This includes utilizing infrastructure as code (IaC) and security as code (SaC) methods to Zero Trust policies and configurations, alongside application code.

By tackling these developments researchers can contribute to advancing Zero Trust Architecture and aiding organizations in keeping up with emerging cybersecurity threats in an evolving digital landscape.

- Conclusion –

In summary, Zero Trust Architecture (ZTA) is a cybersecurity concept that challenges security models by focusing on verification and strict access controls rather than blind trust. This document has discussed the evolution of ZTA tracing its roots back, to the 2000s and its current widespread adoption across sectors. Zta principles like "never trust always verify". Least privilege access has been explained, showcasing the proactive and flexible nature of this strategy.

The report explored the components and framework of ZTA highlighting elements such as micro-segmentation, monitoring, and encryption. Realworld instances in industries like Google BeyondCorp and applications in government and healthcare demonstrated how ZTA enhances cybersecurity defenses and protects assets.

Furthermore discussions on emerging technologies such as Zero Trust Network Access (ZTNA) intelligence, blockchain, and software-defined networking emphasized the progress and research in ZTA. Advancements in areas like IoT native environments, and operational technology integration with DevSecOps aim to improve the effectiveness and scalability of ZTA implementations.

As companies navigate through evolving cyber threats adopting ZTA principles provides a thinking and adaptable approach, to cybersecurity. By focusing on verification limiting access, to necessary privileges, and implementing strict access controls companies can enhance their security position reduce risks, and protect their digital resources in a world that is becoming more interconnected. Ultimately Zero Trust Architecture signifies a change, towards a forward-thinking cybersecurity approach that is wellequipped to tackle the complexities of today's digital age.

References

ferrumgate. (2023, jan 27). *Zero Trust Architecture*. Retrieved from ferrumgate: https://ferrumgate.com/blog/zero-trust-architecture/

Jakkal, V. (2021, June 30). *The critical role of Zero Trust in securing our world*. Retrieved from microsoft: https://www.microsoft.com/en-us/security/blog/2021/06/30/the-critical-role-of-zero-trust-insecuring-our-world/

Key components of a zero trust architecture. (n.d.). Retrieved from aws: https://docs.aws.amazon.com/prescriptive-guidance/latest/strategy-zero-trustarchitecture/components.html

Pratt, M. K. (2022, oct 12). *The history and evolution of zero-trust security*. Retrieved from Mary K.Pratt: https://www.techtarget.com/whatis/feature/History-and-evolution-of-zero-trust-security

Raj, R. (2023, september 29). *Revolutionizing Cybersecurity: The Zero Trust Model*. Retrieved from Linkdin: https://www.linkedin.com/advice/1/what-do-you-want-enhance-cybersecuritypractices-mmwbf

Scott Rose, Oliver Borchert, Stu Mitchell, Sean Connelly. (2020). *Zero Trust Architecture.* National Institute of standards and techology.

TEERAKANOK, P. P. (2023). *A Comprehensive Framework for Migrating.*

tigera. (n.d.). *Zero Trust Architecture: The Basic Building Blocks*. Retrieved from Tigera: https://www.tigera.io/learn/guides/zero-trust/zero-trust-architecture/

What Is Zero Trust? (n.d.). Retrieved from zscaler: https://www.zscaler.com/resources/security-termsglossary/what-is-zero-trust

What Is Zero Trust? Architecture, Principles, and Technology. (n.d.). Retrieved from Tigera: https://www.tigera.io/learn/guides/zero-trust/

wikipedia. (n.d.). *Zero trust security model*. Retrieved from wikipediA: https://en.wikipedia.org/wiki/Zero_trust_security_model